

Find the length of the curve.

1. $y = \left(4 - x^{2/3}\right)^{3/2}$ from $x = 0$ to $x = 8$.

$$y' = -\frac{3}{2}x^{-1/3} \frac{2}{3} \left(4 - x^{2/3}\right)^{1/2} = -\frac{\left(4 - x^{2/3}\right)^{1/2}}{x^{1/3}} \text{ by the chain rule, and}$$

$$(y')^2 = \left(-\frac{\left(4 - x^{2/3}\right)^{1/2}}{x^{1/3}}\right)^2 = \frac{4 - x^{2/3}}{x^{2/3}}.$$

So then

$$\begin{aligned} \int_0^8 \sqrt{1 + (y')^2} \, dx &= \int_0^8 \sqrt{1 + \frac{4 - x^{2/3}}{x^{2/3}}} \, dx \\ &= \int_0^8 \sqrt{\frac{x^{2/3} + 4 - x^{2/3}}{x^{2/3}}} \, dx \\ &= \int_0^8 \sqrt{\frac{4}{x^{2/3}}} \, dx = \int_0^8 2x^{-1/3} \, dx \\ &= \frac{2}{2/3} x^{2/3} \Big|_0^8 = 3x^{2/3} \Big|_0^8 = 12 - 0 = 12 \end{aligned}$$

Find the derivative using logarithmic differentiation.

$$\frac{d}{dx} \ln f(x) = \frac{1}{f(x)} f'(x) \text{ (by the chain rule)} \Rightarrow f'(x) = f(x) \frac{d}{dx} [\ln f(x)]$$

2. $f(x) = x^{\ln x}$

So, $\ln(f(x)) = \ln(x^{\ln x}) = \ln(x) \ln(x)$ ($\ln x^y = y \cdot \ln x$), and

$$\begin{aligned} f'(x) &= x^{\ln x} \frac{d}{dx} [\ln(x^{\ln x})] \\ &= x^{\ln x} \left[\frac{1}{x} \ln(x) + \frac{1}{x} \ln(x) \right] \\ &= \frac{2x^{\ln x} \ln(x)}{x} = 2x^{\ln x - 1} \ln x. \end{aligned}$$

It is usually preferable to combine the variables when possible. This is why the text has the answer in the last form.